

Conversational Voice Interfaces: Translating Research Into Actionable Design

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Abstract

HCI research has for long been dedicated to better and more naturally facilitating information transfer between humans and machines. Unfortunately, humans' most natural form of communication, speech, is also one of the most difficult modalities to be understood by machines – despite, and perhaps, because it is the highest-bandwidth communication channel we possess. As significant research efforts in engineering have been spent on improving machines' ability to understand speech, research is only beginning to make the same improvements in understanding how to appropriately design these speech interfaces to be user-friendly and adoptable. Issues such as variations in error rates when processing speech, and difficulties in learnability and explainability (to name a few), are often in contrast with claims of success from industry. Along with this, designers themselves are making the transition to designing for speech and voice-enabled interfaces. Recent research has demonstrated the struggle for designers to translate their current experiences in graphical user interface design into speech interface design. Research has also noted the lack of any user-centered design principles or consideration for usability or usefulness in the same ways as graphical user interfaces have benefited from heuristic design guidelines.

The goal of this course is to inform the CHI community of the current state of speech and natural language research, to dispel some of the myths surrounding speech-based interaction, as well as to inform participants about currently existing design tools, methods and resources for speech interfaces (and provide hands-on experience with working with them). Through this, we hope that HCI researchers and practitioners will learn how to combine recent advances in speech processing with user-centred principles in designing more usable and useful speech-based interactive systems.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; **Natural language interfaces**.

Keywords

speech interfaces, conversational voice interfaces, speech interface design

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1 Benefits

The course will be beneficial to all HCI researchers or practitioner who still believe in fulfilling HCI's goal of developing methods and systems that allow humans to naturally interact with the ever increasingly ubiquitous mobile technology, but are disappointed with the lack of success in using speech and natural language to achieve this goal. This course will also benefit those who are disappointed with the lack of practical tools that are available to aid and guide designers on how to build usable speech interfaces, and would like to learn about the current advances in design tool development for speech interfaces.

2 Intended Audience(s)

The course is intended for HCI researchers or practitioners who are interested in the development of usable speech interfaces, but feel a lack of foundation directed towards how natural and usable speech interaction can be achieved. This course is also intended for audiences who may or may not be familiar with recent advances in voice and speech interface research, practices, and resources over the last decade, and who are interested in how to translate recent research from the conversational voice interface domain into practical tools that can be used in speech and voice interface design. Even those without a strong expertise in Automatic Speech Recognition (ASR) or Text To Speech (TTS) technologies will find benefit from taking this course.

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3 Prerequisites

There is no specific prerequisites to attend this course, although basic knowledge about speech interfaces may be useful. No technical experience is necessary.

4 Content

Recent advances in Deep Neural Networks have dramatically improved the processing accuracy of speech recognition systems. However, even the most capable computation servers continue to struggle when acoustic, language, or interaction environments are adverse, resulting in large variations in the accuracy of processing speech – this is particularly relevant for home-based smart personal assistants such as Amazon Echo where unexpected interaction contexts (e.g. loud music). This can negatively impact performance and thus user experience. Further discussion and analysis of such examples will be carried out during the course presentation, and in particular, how these challenges affect the design process of conversational voice agents. The course will explore the history of speech interface advancements, and how that has led to the design processes that currently exist in industry, and the current problems that exist in developing such interfaces. This will involve exploring current practices of industry VUI designers, as identified in recent research [4, 6], and the tools that are currently in use [4].

The aim of the presentation is two-fold: present new concepts to the audience, and foster discussions and exchange of ideas. Slides will be used to introduce the main points, while videos and audio clips will be played to illustrate various examples. After (and during) each of the main concepts of the tutorial outline is presented, time will be allocated for interaction between presenters and audience.

New to 2025 is a renewed focus on theoretical and practical review of most recent research on developing design guidelines for conversation user interfaces, as recent research has put efforts to explore [1, 3, 5]. In particular, a new set of guidelines that have been developed and published in 2023, will be presented to participants, representing the most universally proposed principles of VUI design in academic literature, developed by the first-author [2]. The review and exploration of these guidelines is meant to provide a springboard for the adoption and improvement for future tools and resources for speech interface development.

5 Practical work

The course also includes several interactive, hands-on activities. The first activity will engage participants in proposing design alternatives for the error-handling interaction of a smartphone's voice-based search assistant, based on an empirical assessment of the type of ASR errors exhibited (e.g. acoustic, language, semantic). The second activity will center around uncovering speech processing errors of a home-based personal assistant and designing interactions that maintain a positive user experience in the face of unexpected variations in speech processing accuracy. The third activity will center around hands-on practice with the guidelines presented during the presentations, with participants conducting a heuristic evaluation of a readily-available system (e.g. Alexa, Google Assistant, Siri) as guided by these heuristic guidelines. This activity will allow include brainstorming about the benefits and limitations of the guidelines.

6 Instructor background

Christine Murad is a Post-Doctoral fellow in the School of Computer Science at the University of Carleton, and a Research Associate at the University of Waterloo. Her research looks at the usability and design of conversational voice interfaces, the development of different tools and resources to aid designers in CUI & VUI design and education, and bridging the industry-academia gap in CUI & VUI design. She has published research and organized CUI workshops across several international SIGCHI venues such as CHI, IUI, MobileHCI, CSCW, CUI, etc. She has been part of the CUI conference steering committee since its inception in 2019, and has served on the organizing committee for the CUI conference in different track chair roles from 2020 to the present.

Cosmin Munteanu is an Associate Professor and Schlegel Research Chair in Technology for Healthy Aging at the Department of Systems Design Engineering, University of Waterloo, and Director of the Technologies for Ageing Gracefully lab. Cosmin takes a primarily ethnomethodological approach to study how voice- and language-enabled interfaces should be designed in a safe, effective, inclusive, and ethical manner, in order to empower digitally under-represented groups such as older adults. Cosmin is the co-founder of the new ACM Conversational User Interfaces conference series and one of the early researchers to bring speech processing and interface design research together. He is also serving as a member of the ACM SIGCHI Ethics Committee where he contributes perspectives on ethical research with underrepresented user groups.

Gerald Penn is a Professor of Computer Science at the University of Toronto. His area of expertise is in the study of human languages, both from a mathematical and computational perspective. Gerald is one of the leading scholars in Computational Linguistics, with significant contributions to the formal study of natural languages. His publications cover many areas, from Theoretical Linguistics, to Mathematics, and to Automatic Speech Recognition, as well as Human-Computer Interaction.

7 Resources

The course slides will be available to participants at: <http://www.speech-interaction.org/CHI2025course/>. This course proposal is based on conference tutorials and courses by same authors:

- "Conversational Voice Interfaces: Translating Research Into Actionable Design", course presented at CHI 2024
- "Conversational Voice User Interfaces: Connecting Engineering Fundamentals to Design Considerations", course presented at CHI 2021
- "Speech-based Interaction: Myths, Challenges, and Opportunities", course presented at the CHI 2011, 2012, 2013, 2014, 2015, 2016, and 2017, and at HCI 2016, 2017, 2018, 2019, and 2020.
- "Hands-free Interfaces: The Myths, Challenges, and Opportunities of Speech-based Interaction", tutorial presented at MobileHCI 2010, 2011, 2012, 2013, 2014, 2017, and 2018.
- "Natural Interaction for Serious Games: Enhancing Training Simulators through Automatic Speech Recognition", tutorial presented at the I/ITSEC Conference, 2010, 2011, 2012, 2013, 2014, 2015, 2016, and 2017.

The proposed CHI presentation is continuously updated to include more examples and analysis of recent commercial adoption of speech recognition, notably developments on mobile platforms (such as Apple's Siri, Google's Voice Search, or Microsoft's Cortana) or to smart home personal assistants (e.g. Amazon's Alexa). Similar to previous courses, and based on the feedback received from participants, the proposed CHI course material is currently being updated with new interactive examples and short group exercises. The guidelines presented during the workshop will also be provided to participants to keep and use in the future.

8 Accessibility

All materials will be accessible through the course website and available during the course, so that those with hearing or vision impairments may access the content without difficulty. Course instructors will take extra steps to make sure that website materials are accessible for all users by providing meaningful ALT text for images and using colourblind-safe schemes. Hybrid attendance will be accommodated for this course - while there are some activities which may not be possible to be held in a virtual form, we will invite virtual participants to reflect on appropriately discussed topics during the given time.

References

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